



Who Is Ahead?

Quick facilitation notes & answer key

Goal

Students run two concurrent scripts and COMPARE the combined result: which sprite is on the higher number (ahead) and by how many. Comparing the two parallel landings — not just reading one script — is the Grade-2 C3.2 reasoning about how concurrent code behaves together.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters (green for Bit, purple for Pixel) on a floor number path 0-5; after 'green flag!', see whose counter ends further along.

Run it (about 20 min)

1. I DO: read Bit's script — 'start 0, forward 4 → 4, back 1 → 3.' Bit lands on 3.
2. WE DO: read Pixel's script — 'start 3, forward 1 → 4, forward 1 → 5.' Press the green flag so BOTH run together.
3. YOU DO: Exercise 3 — students compare: Pixel (5) is ahead of Bit (3) by 2 numbers.
4. Connect: 'Because both ran at the same time, we can line them up and see who is ahead.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 4 → 4 → back 1 → 3. Bit lands on 3.

Ex 2 — Pixel: 3 → forward 1 → 4 → forward 1 → 5. Pixel lands on 5.

Ex 3 — both start on the green flag; Pixel (5) is on the higher number, so Pixel is AHEAD, by $5 - 3 = 2$ numbers.

Interaction note: the two scripts run at the same time and independently — we compare their final positions to decide who is ahead.

Watch for & differentiate

Common error: thinking the sprite with more moves is automatically ahead — stress that 'ahead' means the higher landing NUMBER, not the longer script.

Common error: running the scripts one after the other instead of together — the green flag starts BOTH at once (concurrent).

Simplify: find each landing first, then just compare the two numbers (which is bigger?).

Extend: ask what move you could add to Bit's script so the two sprites would TIE.

Success indicator: the child lands Bit on 3 and Pixel on 5, names Pixel as ahead, and says the gap is 2.